

EM 547

MANAGEMENT OF RESEARCH DEVELOPMENT

COURSE DESCRIPTION

The purpose of this course is to introduce empirical approaches to assessing critical issues within the field of Engineering Management.. The course will simultaneously present various methodological and statistical approaches for conducting research on various technological issues within society, the community, and formal organizations.

COURSE OBJECTIVES

Upon completion of this course, students will be able to:

- Identify technological trends and develop a research problem that addresses the nature of the relationships between a specific technology and societal/organizational outcomes
- Compare and contrast the differences between longitudinal, cross sectional, and experimental research designs.
- Develop, organize, and implement an empirical research study relevant to technology studies
- Understand and utilize explanatory statistics including analyses of variance, correlation, and cross-tabular analyses.
- Understand and utilize descriptive statistics including histograms, z scores, measures of central tendency.
- Utilize the Statistical software (Excel or SPSS or MiniTab) for analyzing quantitative data.
- Evaluate, review and critique empirical research studies presented within professional journals.

COURSE OUTLINE AND SCHEDULE

Unit #1

Introduction

Chapter 1 and Chapter 2

Unit#2

Chapter 3 and Chapter 4

Group Research Project Assignments Due

Unit#3

Chapter #5 and Chapter #6

Group Research Project Assignments Due

Unit#4

Chapter #7 and Chapter #8

Group Research Project Assignments Due

Unit#5

Chapter #9 and Chapter #10

Unit#6

Chapter 11

Unit#7

Chapter 12

Unit#8

Final Exam

Research Report Submission

EXPLANATION OF COURSE CRITERIA

UNIT EXAMS: These exams will consist of a combination of true/false, multiple choice, and essay questions. Each exam will only cover the material within the designated units.

HOMEWORK ASSIGNMENTS: Most of the homework assignments will consist of statistical procedures utilizing SPSS or your preferred statistical software. However, several of the assignments will also contain such things as literature reviews.

THREADED DISCUSSIONS: The threaded discussions (TD) must be taken as seriously as any other course assignment. These discussions are representative of an actual face-to-face classroom setting. This means that you must post your ideas within the time frame stipulated within the TD discussion assignment. Postings submitted beyond the deadline will be penalized.

The grading of the TD will be based upon the following criteria:

1. The extent to which the TD assignment was completely addressed.
2. Responding to at least one or two other student postings
3. The correct use of grammar and spelling
4. Rubric provided

GROUP RESEARCH PROJECT: The Group project is team effort and should be treated as such. You will start by creating teams and then working of a research based on the criteria provide in initial units. It is advisable to complete your work on time by starting the project on time. Follow Research Paper outline format and rubric to obtain maximum marks. The grading of the group research project will be based on the following criteria:

1. The comprehensiveness of the literature review
2. The extent to which the ordinal variables have been theoretical defined
3. The extent to which the scales representing the ordinal variables are reflective of the theoretical (operational definitions) definitions of each of the ordinal variables.
4. The proper utilization of the statistical procedures and analysis of the data
5. The extent to which the structure of the submitted group projected follows the stipulated research format

IMPORTANT NOTE: All assignments must be submitted by the stipulated due dates. Late assignments will have points deducted unless an extenuating circumstance describing why an assignment cannot be submitted on time is presented to the instructor. Please do no wait until the last moment to contact the instructor concerning such circumstances.

You may use email or phone to consult with the instructor as well as make a face-to-face visit on Skype during the instructor's designated office hours.

WEIGHTAGE:

Unit Exams:	20%
Threaded Discussions:	10%
Individual Assignments:	20%
Team Assignments:	20%

Research Project:

30%

ADDITIONAL LITERATURE

Nurusis, Marija. (2005). *SPSS Guide to Data Analysis*. New Jersey. Prentice-hall

Denzin, Norman. (1999). *The Research Act*. 3rd Edition. .New Jersey. Prentice-Hall.

Sirkin, Mark K. (1995) *Statistics For the Social Sciences*. Thousand Oaks. Sage Publications.

SPECIFIC COURSE POLICIES, GUIDELINES, AND GRADING

A. Access to Course.

Our course consists of 8 units, which are scheduled from Mondays through Sundays. Most units will be scheduled across two full weeks with the exception of Unit 1, which starts on a Wednesday, and Unit 8, which ends on a Friday. **All assignments will be due at the end of each unit on Sunday nights, by 11:59 PM ET, except for selection of presentation topics in Unit 1 (due earlier), and the submission of formal papers, which must be submitted by the Sunday night that is one week prior to the opening of unit of that presentation to the class. No assignments required to be completed for a unit will be accepted after the close of that unit.**

All completed units will remain open for review, until the end of the semester.

B. Email to Instructor.

For EM 695 course-related submissions and communications, please send all email directly to Dr. Ahmed at his personal email account, at mahmed6@emich.edu. Also, in the subject line of your email messages, include the: **initials of your full name**, **course number**, and the **subject of the message** (such as the name of an assignment that is attached to your email); for example, Subject: **XYZ547Design1**. I can not be on call for your class 24 hours a day, but will be checking in as often as possible. I expect to check EM 547-related email before 12 noon on several days during the week. Due the amount of email I will receive from students, as well as other work and travel commitments, I may not be able to respond to emails or confirm receipt of submitted assignments more than once or twice a week--although I will try to do so more frequently. So, plan accordingly. If you wait until the last minute to do an assignment, and then email a request for assistance with a problem, neither I nor a fellow student may be able to respond quickly enough to help you. Also, it make take a day or two before you receive an emailed confirmation from me that I received your submitted assignment. Typically, the confirmation will read: "Got it, Dr. Sohail."

C. Assignment Software, Submission to Drop Box, and File Names.

There are so many different word processing programs that reading assignments can become a problem in online courses if there is no standard policy set for them. **It is preferred that you submit papers in MS Word file format.** Formats that are compatible with MS Word include earlier Word versions, Microsoft Works, Word Perfect 5 or 6, and Rich Text Format. If necessary, you can compose the work in your non-compatible word processor and simply cut and paste it into an email message--although this may cause a problem if formatting is important. If using MS Project software, please submit your work in MS Project 2007 format. You may also send single or multiple files in .zip compressed format.

Submit assignments to the Unit Drop Box as directed for each Unit. The file names used for assignments should follow a coding format similar to that used for course-related emails. For example, include the: initials of your full name, course number, and the name of the assignment with its file type; therefore, the file name of an assignment would look like: XYZ547Presentation1.doc.

I will use TurnItIn.com to check papers for originality after they are submitted to the drop box.

D. Assignment Delivery and Technical Problems.

Since you will have more than two weeks and three weekends available for completing your work throughout most of the course, we expect you to get it done on time. **No assignments will be accepted after the close of a unit.** Written paper assignments are to be submitted using the Unit Assignment Drop Box, and are due one week prior to the close of a unit (see schedule in syllabus). Written paper assignments will not be accepted after the close of a unit. Other work such as threaded discussions, quizzes, and exams are submitted using those course tools, which automatically close at the end of a unit. The final exam (and any other assignments due in our Unit 8 final week) will not be accepted late. We will provide early access to units for you. Plan on getting your work done during the first half of the unit, and plan on encountering unexpected problems. **You have been made aware of the consequences of poor planning, procrastination, and not getting your work done on time.**

In some cases your computer may freeze or you might be having issues with the eCourse or eCompanion web site (your exam or quiz or assignment not saved or your being kicked out from the page etc), in all such cases contact eCourse/eCompanion help desk and copy your instructor on all such emails. If help desk confirms that the issues was related to them then you instructor will give you extra time to finish all such assignments. So please make sure that you don't wait until last day or last hour to finish up your assignments.

We encourage you to make hard copies of your assignments and communications with your instructors. Computer crashes and disasters can happen--we expect you

to practice good risk management behavior, just like good engineering managers. We do not view ISP problems as an excuse for turning work in late. Alternative methods of submitting assignments include:

1. Fax it to Dr. Sohail at 734.487.8755 (EMU).
2. Overnight express hard copy and on disk.
3. Make arrangements over the phone with me.

Note: It is your responsibility to access the course on another computer in the event of hardware or software problems. This can be done with a friend's computer, or one at Kinko's or the Public Library. Our guidance from the university is to not consider any excuse for missing assignments due to technical difficulties as legitimate.

E. Quizzes and Exams.

Most of the quizzes and exams in this course are timed, based on an easy standard of 120 seconds (two minutes) per question typically. The typical standard for professional-level exams, such as certification or licensure exams, is about 60 to 90 seconds per question. What makes it even easier for you, is that your quizzes and exams in our course are very focused on specific information from sections of specific books--whereas certification and licensure exams cover a very wide body of professional knowledge, covered by numerous books and other material.

Like some professional exams, your quizzes and exams in this class are open-book (of course)--and, like those professional exams (and regular classroom exams), you should prepare for taking your quizzes and exams in advance, by reading the material to be covered, taking notes, and applying other test-preparation studying techniques.

F. Evaluation and Grading.

Course grades will be determined by the percentage of total points attained for various work assignments completed throughout the semester, based upon the following scale:

100 to 93% = A

92.99 to 90% = A-

89.99 to 87% = B+

86.99 to 83% = B

82.99 to 80% = B-

79.99 to 77% = C+

76.99 to 73% = C

72.99 to 70% = C-

(no D grades, based on Grad School requirements)

69.99 and below = E

Final course grades will be based primarily (*but not necessarily only*) on the quantitative scale shown above--but are subject to adjustment based on the judgment of your professor in consideration of qualitative factors, including academic dishonesty and displays of other unethical, unprofessional, or uncivil behaviors that

are unbecoming of a good engineering manager and graduate student. (*You must to be prepared for the reality of professional life in business and industry--there is usually more to getting a good performance evaluation from your boss and peers than just production of work, other factors include: (a) personal professional skills, (b) professional communication skills, and (c) ethics and emotional intelligence.*) (a) (b) and (c) above are of vital importance to becoming successful as an Engineering Manager, and why they are three of our EM Program Student Learning Objectives. We expect to see them demonstrated by students in our classes.

G. Personal Involvement and Time.

Students often have the misperception that taking a course online is easier since they don't have to physically go to a classroom. When taking an online course there are some things that become easier for students. For example, it becomes easier NOT to go to class. It also becomes easier to get behind. This medium gives students flexibility in attending class--but remember, while taking a course online may be more convenient, it is not necessarily easier!

Be aware of the general guidelines to professors for student work loads: 45 hours of work per credit hour per semester. This is a 3 credit class, so we expect, on average, that students will spend 135 hours of total work time in this course. Students with more graduate-level and online learning experience tend to be more efficient and require less time--some students will require a great deal more time to get their school work completed. Be prepared for the work involved, as it may require 10-15 (or more) hours of your time each week.

This is an Engineering Management course, part of the EM program, taken by EM masters and doctoral students. Engineering Managers are responsible for planning, organizing, staffing, leading, and controlling engineering projects and personnel. They must lead by example, and make sure that objectives are met and completed on time. Poor planning and procrastination, such as trying to complete all work at the last minute, leads to poor quality and missed deadlines because there is no buffer available to deal with unexpected problems. Many of our EM students lead very busy professional lives and have significant work and family responsibilities in addition to going to school--yet they do their school work well, and get it done on time (often early). If a student in our program cannot organize themselves well enough to get their school work done on time, then they will certainly have a problem organizing and leading other engineers and engineering projects, and probably should not be in this program. As such, we will give little leeway when it comes to assignments that are not submitted on time--and will suggest to chronically late students that they seek a different degree and profession.